

Alireza Alinejad

✉ Alireza.Alinejad.g.f.a@gmail.com | 📞 +98 919 8020208 | 🌐 Alireza-Alinejad-2002.github.io

📍 Tehran, Iran | 🔗 linkedin.com/in/2002-alireza-alinejad | 🏠 github.com/Alireza-Alinejad-2002

Objective

I believe that as AI models continue to evolve, we are closer than ever to achieving a human-like robotic brain. However, this brain needs an advanced, adaptable body to execute actions that genuinely ease human lives. I am driven to explore the creation of such robots, merging intelligent design with purposeful functionality to bridge the Moravec's Paradox gap.

Education

Sharif University of Technology (SUT), B.Sc. in Aerospace Engineering

Sep 2021 – Present

• GPA: 3.67 (17.36 out of 20)

[Original Transcript]

Areas of Interest



Robotics



Machine Learning



Control & Autonomy



Computer Vision



Swarm Intelligence



Mechatronics

Patents and Publications

C=Conference, J=Journal, P=Patent, S=In Submission, T=Thesis

[P.1] A. Alinejad, T. Alikhani, S. H. Pourtakdoust, S. M. S. Mousavi. (2025). **Design and Development of a Dragonfly-Inspired Flapping Mechanism**. Iran Intellectual Property Office, Application No. 140450140003008393 (In Submission).

Research Experience

• Researcher at Smart Systems Research Laboratory

↪ SSR Lab, Aerospace Engineering Department, SUT

Aug 2023 – Present

Supervisor: **Prof. Seyed Hossein Pourtakdoust**

• Dragonfly-Inspired Flapping Wing Micro Air Vehicle (FWMAV)

↪ Tools: SolidWorks, Rapid Prototyping (mostly FDM), Arduino, MATLAB, Python

Aug 2023 – Sep 2025



- Designed biomimetic flapping mechanism via more than 4 iterative prototyping.
- Integrated control system for enhanced maneuverability and efficiency.
- Validated performance through simulations and physical tests.

• Inverted Rotary Pendulum Control with Raspberry Pi

↪ Tools: Raspberry Pi, SSH Protocol, C++

Sep 2025

- Configured Raspberry Pi via SSH for control experiments.
- Implemented stabilization algorithm for rotary pendulum.
- Tested system performance in self-directed project.

Skills

- **Soft Skills:** Problem-Solving, Communication, Teamwork, Critical Thinking
- **Programming Languages:** Python, C++, MATLAB, Simulink, Arduino
- **Data Science & Machine Learning:** Python, TensorFlow, MATLAB
- **Mathematical & Statistical Tools:** MATLAB, EES (Engineering Equation Solver)
- **Other Tools & Technologies:** ROS2, SOLIDWORKS, Linux, $\LaTeX$
- **Research Skills:** Data Analysis, Statistical Modeling, Literature Review, Experimental Design, Qualitative Research, Quantitative Research

Language Proficiency

- **English** ↔ TOEFL iBT: 97 (R24, L28, S24, W21)
- **Persian** ↔ Native

Academic Experience

- **Aircraft Design, Teaching Assistant**
↪ Sharif University of Technology (SUT)
Sep 2025 – Present
Instructor: **Prof. Afshin Banazadeh**
- **Numerical Methods, Teaching Assistant**
↪ Sharif University of Technology (SUT)
Sep 2025 – Present
Instructor: **Dr. Hossein Hashemi Nasab**
- **Control Systems Lab, Teaching Assistant**
↪ Sharif University of Technology (SUT)
Sep 2024 – Jan 2025
Instructor: **Prof. Alireza Sharifi**
- **Engineering Dynamics, Teaching Assistant**
↪ Sharif University of Technology (SUT)
Sep 2023 – Jan 2024
Instructor: **Prof. Alireza Sharifi**

Selected Course Projects

- **ROS2-Based Mobile Robot Navigation and NAO Robot Interaction**
↪ Tools: ROS2, Gazebo, TurtleBot3, NAO Robot, CHOREGRAPHE
Feb 2025 – Jun 2025
 - Simulated TurtleBot3 for environment mapping in Gazebo.
 - Deployed code on real robot for autonomous exploration.
 - Programmed NAO for movement and interaction tasks.
- **Reimplementation of MCST Tracking Algorithm**
↪ Tools: Python, Bi-LSTM Networks, PyTorch
Feb 2025 – Jun 2025
 - Reproduced adaptive tracking model for high-speed targets.
 - Incorporated Bi-LSTM and maneuver compensation unit.
- **Hybrid-Electric STOL Air Taxi Design (DEP System)**
↪ Tools: MATLAB, OpenVSP, SolidWorks, CS-23 Regulations
Feb 2025 – Jun 2025
 - Engineered hybrid-electric aircraft featuring 10-motor Distributed Electric Propulsion.
 - Conducted full weight and performance sizing, validated aerodynamics via CFD, and optimized powertrain.
 - Designed interior configuration meeting CS-23 airworthiness standards.
- **Amphibious Aircraft Design (RAeS 2024-2025 RfP)**
↪ Tools: MATLAB, SolidWorks, Analytical Modeling, CS-23 Regulations
Sep 2024 – Jan 2025
 - Developed conceptual design meeting CS-23 standards.
 - Conducted sensitivity and performance analyses.
 - Optimized aerodynamics for operational versatility.
- **Quadrotor Control System Simulation**
↪ Tools: MATLAB, Simulink, SolidWorks, Simscape
Feb 2024 – Jun 2024
 - Modeled 6DOF dynamics with PID control strategy.
 - Simulated system for stability validation.
 - Aligned results with robotics precision standards.
- **Airline Performance Optimization**
↪ Tools: Python, Data Analysis, Web Scraping
Sep 2023 – Jan 2024
 - Integrated multi-source metrics via data analytics.
 - Developed predictive models for decision support.
 - Automated web scraping for insights.

Honors and Awards

- **Top Rank at Aerospace Engineering Department** Sep 2021 – Present
↪ Aerospace Engineering Department, SUT
 - Ranked 17 among 78 students.
 - Studying Aerospace Engineering at 1st rank Engineer University in Iran, SUT.
- **Elite National Rank** Jul 2021
↪ Iranian Nationwide University Entrance Exam
 - Ranked among the top **0.3%** of participants in the Iranian Nationwide University Entrance Exam.

Professional Memberships

- **Head of Welfare Committee** May 2023 – Sep 2024
↪ Aerospace Engineering Department's Student Council, SUT
 - Led the Welfare Committee, overseeing student well-being initiatives.
 - Developed and implemented programs improving student support and resources.
 - Honed leadership, organizational, and communication skills.
- **Environmental Cooperation** Apr 2023 – Dec 2023
↪ Association of Environmentalists, SUT
 - I have been associated with multiple programs such as tree-planting, paper waste recycling, etc.
- **Technical Editor** Oct 2022 – Apr 2023
↪ Aerospace-Based Trade Magazine, OWJ Publications, SUT
 - Edited multiple magazines for grammatical and scientific accuracy.

Certifications

- **Deep Learning Fundamentals: Unlocking the Power of AI** Aug 2025
- **Artificial Neural Network and Machine Learning using MATLAB** Jul 2024
- **Simulink Onramp** Mar 2024
- **MATLAB Onramp** Nov 2022

References

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| <ul style="list-style-type: none">• Seyed Hossein Pourtakdoust
<i>Professor, Faculty of Aerospace Engr Dept</i>
Sharif University of Technology, Tehran, Iran
Email: pourtak@sharif.edu
Phone: +98-21-66168163
<i>Relationship: B.Sc. Team Project Supervisor, Course Instructor</i>• Alireza Sharifi
<i>Assistant Professor, Faculty of Aerospace Engr Dept</i>
Sharif University of Technology, Tehran, Iran
Email: ar.sharifi@sharif.edu
Phone: +98-21-66168115
<i>Relationship: Teaching Assistant, Course Instructor</i>• Abbas Ebrahimi
<i>Associate Professor, Faculty of Aerospace Engr Dept</i>
Sharif University of Technology, Tehran, Iran
Email: ebrahimi_a@sharif.edu
Phone: +98-21-66166201
<i>Relationship: Course Instructor</i> | <ul style="list-style-type: none">• Afshin Banazadeh
<i>Professor, Faculty of Aerospace Engr Dept</i>
Sharif University of Technology, Tehran, Iran
Email: banazadeh@sharif.edu
Phone: +98-21-66168108
<i>Relationship: Teaching Assistant, Aircraft Design Team Project Supervisor, Course Instructor</i>• Hossein Mohammad Navazi
<i>Associate Professor, Faculty of Aerospace Engr Dept</i>
Sharif University of Technology, Tehran, Iran
Email: navazi@sharif.edu
Phone: +98-21-66168175
<i>Relationship: Course Instructor</i>• Azadeh Kebriaee
<i>Assistant Professor, Faculty of Aerospace Engr Dept</i>
Sharif University of Technology, Tehran, Iran
Email: kebriaee@sharif.edu
Phone: +98-21-66164600
<i>Relationship: Course Instructor</i> |
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